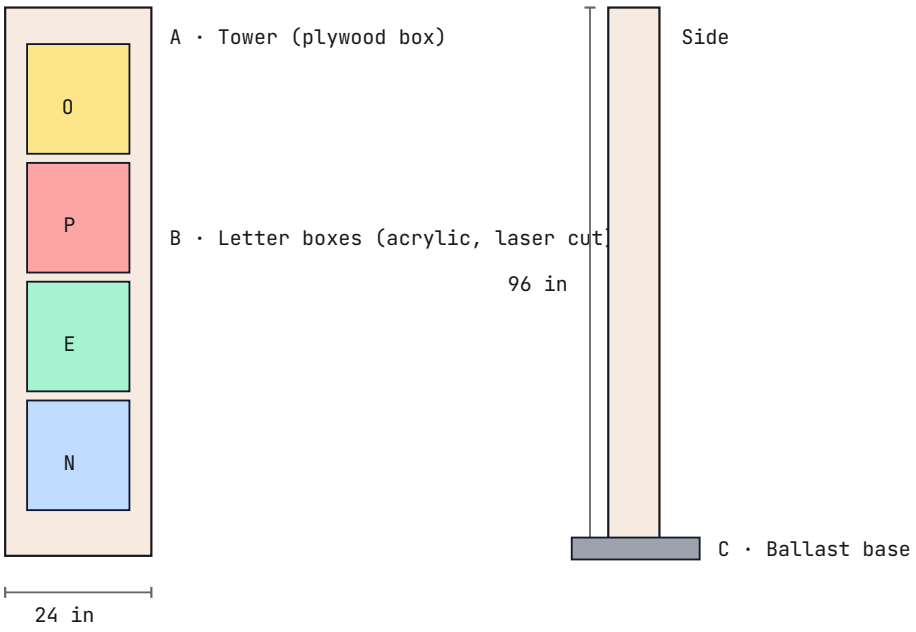
BUILD SPEC · NOVEMBER 2026 · SAMPLE, FICTIONAL PROJECT

Night Market sign

An 8 ft freestanding sign that spells OPEN LATE in color-changing letters, for an outdoor night market, running all night on one battery.

2 × 8 × 0.7 ft	$1,205	294 W	4 days
OVERALL SIZE	MATERIALS	POWER	BUILD TIME
	estimate incl. tax, Nov 2026	10.4 h on a 3.6 kWh battery	2 people

01 Overview



A · Tower	1/2 in birch plywood box on a 2×2 frame, track-saw cut, glued and screwed
B · Letters	Eight laser-cut acrylic boxes (white face, black returns) with LED strip inside
C · Base	Plywood plinth with four sandbags and ratchet straps to a stake or fence

02 Bill of materials

PART	QTY	UNIT	EACH	TOTAL	SOURCE	NOTES
Structure						
1/2 in birch plywood 4×8	3	sheet	$68.50	$205.50	link	cabinet grade
2×2 poplar 8 ft	6	piece	$9.80	$58.80	link	inner frame

PART	QTY	UNIT	EACH	TOTAL	SOURCE	NOTES
Face						
1/4 in white acrylic 24×18	7	sheet	$34.00	$238.00	local plastics supplier	letter diffusers
1/8 in black acrylic 24×18	7	sheet	$22.00	$154.00	local plastics supplier	letter returns
Hardware						
#8 × 1-1/4 in wood screws (100)	1	box	$9.99	$9.99		
Ratchet straps and sandbags	4	set	$18.00	$72.00		wind ballast
Electrical						
WS2815 LED strip 5 m 12 V	4	reel	$38.00	$152.00		per-pixel color
Pixel controller (Wi-Fi)	1	each	$60.00	$60.00		
12 V 30 A power supply (weatherproof)	1	each	$45.00	$45.00		
Inline fuse holder + 30 A fuses	2	each	$7.50	$15.00		one per supply output
14 AWG two-conductor wire 50 ft	1	spool	$32.00	$32.00		
Finish						
Exterior primer and black satin paint	1	kit	$58.00	$58.00		two coats
Materials subtotal				**$1,100.29**		
Tax (9.5%)				$104.53		
Total				**$1,204.82**		

03 Cut list and sheets

PART	QTY	SIZE (IN)	MATERIAL	PROCESS
Side panel	2	8 × 96	1/2 in ply	Track saw
Back panel	2	24 × 96	1/2 in ply	Track saw
Letter face	8	16 × 20	1/4 in white acrylic	Laser
Letter return	8	16 × 20	1/8 in black acrylic	Laser

Sheet goods

MATERIAL	SHEET SIZE (IN)	PART AREA (SQ FT)	SHEETS (+15% WASTE)
1/2 ply	48 × 96	44.0	2

MATERIAL	SHEET SIZE (IN)	PART AREA (SQ FT)	SHEETS (+15% WASTE)
1/4 white acrylic	24 × 18	17.8	7
1/8 black acrylic	24 × 18	17.8	7

Sheet counts come from part area plus a waste allowance; real nesting can need one more sheet. Check the layout in your nesting or laser software.

04 Files and machine settings

PROCESS	FILES	MATERIAL	STARTING SETTINGS	MACHINE TIME
Laser	letters.svg: red = cut, blue = score	1/4 in cast acrylic	Cut 100% power, 8 mm/s, 2 passes TEST ON SCRAP	3.5 h
Laser	returns.svg	1/8 in cast acrylic	Cut 90% power, 14 mm/s TEST ON SCRAP	2 h

05 Power

Power budget

DEVICE	QTY	WATTS EACH	WATTS
WS2815 LED strip 5 m 12 V	4	72	288
Pixel controller (Wi-Fi)	1	6	6
Total draw			294 W
With 25% headroom			**368 W** · 30.6 A at 12 V
Runtime on 3,600 Wh battery (85% usable, at total draw)			10.4 h

12 V DC throughout. 30 A supply fused at the output, 14 AWG runs under 15 ft, controller in the base behind a removable panel.

06 Build steps

1. **Tower:** cut panels, glue and screw the 2×2 frame.
   Check: square within 1/16 in before the back goes on.
2. **Letters:** laser the faces and returns, bond with acrylic cement.
   Check: dry-fit every letter in its cutout.
3. **LEDs:** lay strip inside each letter box, solder joins, heat-shrink.
   Check: test every letter before sealing.
4. **Finish:** prime and paint the tower, two coats.
   Check: 24 h cure before install.

07 Safety

Before you build

- Cut acrylic only: never laser PVC or vinyl. Run the laser with ventilation and stay with it.
- An 8 ft sign catches wind: ballast with at least 100 lb and strap it to a fixed point; take it down above 25 mph.
- Fuse the battery output and keep all wiring 12 V DC.

Made with the free Build Spec skill for Claude, by Beautiful Engine · beautifulengine.com